Reproducibility Report: Barari, Coppock, Graham and Padgett (2024)

Did Trump's Indictments Rally His Base? Evidence from the Counterfactual Format

Table of contents

1 Paper Overview 3

2 Summary 3

2.1 Does the deposited archive run? . . . 4

2.2 Does the maintained rewrite reproduce the paper? . . . 5

3 Original Archive Reproducibility 6

4 Ground Truth Verification 8

4.1 In-text claims . . . 11

4.2 Errata . . . 11

5 Maintained Rewrite 11

6 Figure Verification 13

7 R Environment 15

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This repository holds the actively maintained replication code for Barari, Coppock, Graham and Padgett (2024), together with the reproducibility report that documents what the original archive does and does not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article 10.1093/poq/nfae061

Replication archive	10.7910/DVN/QDOBJF
Pre-analysis plan	None registered

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the Dataverse file identifiers, the UNF of the one ingested data file, and two checksums per file: the MD5 of the bytes Dataverse serves, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all twelve files. They do not always, so the script verifies against the served bytes and reports any disagreement. Either way the exact bytes are pinned in version control even though the bytes themselves are not. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it, and also holds `published_claims.csv`, the list of every numeric token in the article and its appendix, and `run_deposited_scripts.R`, which runs the archive's own scripts in a throwaway copy and records both whether they finish and what they compute. `maintained/in_text_claims.R` pairs each of those claims with the sentence that makes it and the number this pipeline produces. `errata.qmd` at the repository root builds `barari_etal_2024_errata.pdf`, a short note on the seven errors the article carries: four places where it states something its own tables and data do not support, and three typographic. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `barari_etal_2024.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies its twelve checksums, reshapes the data, produces every published table and figure into `maintained/output/`, runs the deposited scripts in a scratch copy, and rebuilds the ground truth from all of it. A first run takes a few minutes, most of it downloading the 52 MB deposit. Individual scripts can be run on their own in any order once `maintained/clean_data.R` has run once.

Required packages: tidyverse, estimatr, ggh4x, here for the rewrite itself; xtable and kableExtra in addition, because `run_all.R` also runs the deposited scripts and those load them; knitr and kableExtra to render this report. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. A

successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV and PNG output should come back byte-identical. The six PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Paper Overview

Citation: Barari, S., Coppock, A., Graham, M. H., and Padgett, Z. (2024). Did Trump's indictments rally his base? Evidence from the counterfactual format. *Public Opinion Quarterly*, 88(4), 1216-1233.

Research question: When a major news event has already reached almost everyone, how should a pollster measure its effect on opinion? The article compares two ways of asking respondents to assess that effect themselves.

Design: An opinion poll of 4,838 respondents in the deposited data, fielded with SurveyMonkey between 22 and 27 June 2023, weighted by multistage raking to 2019 American Community Survey targets. Half the respondents were randomly assigned to the *change format*, which asks directly whether the indictment made them more or less likely to hold a view. The other half received the *counterfactual format*, which asks for the current opinion on a 0 to 100 scale and then for the opinion the respondent would have held had they never heard about the indictment. The difference between the two answers is that respondent's own estimate of the event's effect. Three topics: belief that Trump mishandled classified documents, Republican primary vote intention, and general election vote intention. All estimates are weighted means and weighted differences in means with robust standard errors.

Main finding: The two formats disagree in direction. Under the change format, 43 percent of Republican primary voters say the indictment made them more likely to support Trump against 16 percent saying less likely, implying the indictment helped him. Under the counterfactual format the same voters put their chance of supporting Trump at 64.1 percent, against 65.7 percent had the indictment not happened, an estimated effect of -1.6 percentage points. The authors read the gap as response substitution: asked how an event changed them, partisans report the level of their loyalty instead.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Downloaded whole and run as it stands, four of its six scripts run clean and two fail, both figure scripts, for the same reason in the same place. Run instead against a copy holding only the data and the code, five of the six fail.

Table 1: The deposited scripts in a current R session

Script	As deposited	Data plus code only	Issue
prep_data.R	Runs	Runs	Clean
figure_1.R	FAILS	FAILS	ggsave() to C:/Users/tuq69844/Dropbox/... , a path on an author's own machine
figure_A1.R	FAILS	FAILS	ggsave() to C:/Users/tuq69844/Dropbox/... , a path on an author's own machine
table_2.R	Runs	FAILS	Reads an .rds the archive ships but no runnable line of it writes
table_A2.R	Runs	FAILS	Reads an .rds the archive ships but no runnable line of it writes
table_A3.R	Runs	FAILS	Reads an .rds the archive ships but no runnable line of it writes

`figure_1.R` and `figure_A1.R` each end with three `ggsave()` calls writing EPS files to a Windows directory that exists on no other machine. Every line of computation in both scripts completes before the first of those calls is reached, so the failure costs nothing but the figures themselves: running the scripts under `try()` and then writing out the plotting object recovers every plotted value.

Two further facts about the deposit matter more than the failures.

Nothing the archive computes is ever written to a file. Every export call in all six scripts is commented out: the three `write_rds()` lines at the end of `prep_data.R`, the `write()` calls in `table_2.R`, `table_A2.R` and `table_A3.R`, and the local-path `ggsave()` lines in the figure scripts that survive only in their Windows form. Run the archive as deposited and it leaves the directory exactly as it found it. The tables reach the console as LaTeX and nowhere else.

That is also why the four clean scripts are less clean than they look. `prep_data.R` is the script that builds `dat_clean.rds`, `dat_long_clean.rds` and `long_topic_clean.rds`, and those are the three files the other five scripts load; because its `write_rds()` calls are commented out, it builds them in memory and discards them. The five downstream scripts run only because the deposit separately ships finished copies of the same three objects. Take the copies away and

every one of the five stops on the first line that reads one, which is what the third column of the table above records. The deposit is internally reproducible only in the sense that its inputs and its outputs happen to sit in the same folder.

One published table has no script at all. Table A.1, the weighted and unweighted demographic profile of the sample, is not produced by anything in the deposit. Its inputs are present in `data.csv`, so the rewrite rebuilds it; see below for what happens when it does.

2.2 Does the maintained rewrite reproduce the paper?

Every published number with a script behind it reproduces exactly. The table without a script does not, and the reason implicates the deposit rather than the code.

Table 2: Reproduction verdict by component

Component	Verdict
Table 2 (main results)	Reproduces, all 54 checkable cells
Table A.2 (magnitudes)	Reproduces, all 84 cells
Table A.3 (by party identity)	Reproduces, all 84 checkable cells
Figure 1	Reproduces; the article's own text transposes its panel letters
Figure A.1	Reproduces
Table A.1 (sample profile)	Does not reproduce: 35 of 38 cells miss
In-text quantities	Reproduce except where the text contradicts Table 2

Of 283 ground truth claims that can be checked against the rewrite, 236 match the published value and 47 do not. All 222 checkable cells in Tables 2, A.2 and A.3 agree to the tenth of a percentage point the article prints, and so do all 204 of those cells that the deposited scripts also produce. The failures fall into two groups.

The deposited data are smaller than the survey the article describes. The article reports that 5,011 of 6,877 respondents completed the survey. `data.csv` holds 4,838 rows, with ids running 1 to 4,838 and no gaps, and neither `README.txt` nor `codebook.pdf` mentions an exclusion. The 173 missing respondents are invisible in Tables 2, A.2 and A.3, every cell of which reproduces from the deposited data, so the analysis sample is 4,838; the three Ns printed in Figure 1's caption are the deposit's own party identification counts exactly, which is independent evidence for the same figure. They are not invisible in Table A.1, which is a profile of the sample rather than an estimate from it: 35 of its 38 cells are off, by 0.3 of a point at the median and by no more than 1.4. The article's weighted Biden approval check, quoted at 40.4 percent against a FiveThirtyEight average of 40.3, comes out at 41.4 percent on the

deposited data. Nothing here changes a conclusion; the point is that a reader who downloads the deposit cannot rebuild the sample the article describes.

Three passages in the article contradict the article. All three are internal, in the sense that the deposit and the rewrite agree with each other and with the printed tables, and only the surrounding prose disagrees.

- Figure 1's three panels are lettered (a) neither primary, (b) Democratic primary, (c) Republican primary in the printed figure, which is the order `figure_1.R` writes them in. The caption and the body text both assign (b) to Republican primary voters and (c) to Democratic. The printed panels settle it: panel (b)'s belief column shows 79.5 percent saying "more likely" against 4.9 percent "less likely" and its vote column is the general election, which is the Democratic profile; panel (c) carries the Republican primary column and a belief split of 15.1 against 42.7. Those are the unweighted shares Figure 1 plots; Table A.2 gives the weighted versions of the same quantities.
- Figure 1's caption gives the three panel sizes as 754, 2,026 and 1,953. Those are the counts of Independents, Republican identifiers and Democratic identifiers, which are the groups in Figure A.1. Figure 1 splits by intended primary, and those groups number 889, 2,028 and 1,921.
- The paragraph describing respondents who intend to vote in neither primary reports an effect on beliefs of "+1.8 pp, s.e. = 0.9, p = 0.05" and an effect on vote choice of "+0.1 pp, s.e. = 0.9, p = 0.89". Table 2 reports 1.5 with p = 0.124 and 0.3 with p = 0.752 for the same two cells, and the data give the table's values. The same paragraph gives p = 0.02 for the primary vote effect that Table 2 puts at p = 0.007. The standard errors quoted in the text are right in every case, so the sample is not the difference; the estimates and p-values read like a stale draft.

None of the three touches a substantive claim. The neither-primary effect on beliefs is small and near the conventional threshold either way, and the sentence's own characterisation, that the counterfactual format "suggests indifference" for this group, is if anything better supported by Table 2's numbers than by its own.

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3 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/QDOBJF, twelve files, 52 MB.

The deposit contains six R scripts (`prep_data.R`, `figure_1.R`, `figure_A1.R`, `table_2.R`, `table_A2.R`, `table_A3.R`), the respondent-level `data.csv`, three reshaped `.rds` copies of it, a codebook and a README. Scripts load their data by bare filename and must be run from the archive directory. They are run here in a throwaway copy rather than in `original/` itself, because a script that writes into the directory it reads from can overwrite a deposited file.

The two failures. `figure_1.R` and `figure_A1.R` both end with

```
ggsave("C:/Users/tuq69844/Dropbox/indictment_counterfactual/submissions/POQ_final/Barari-23-0
```

and ggplot2 stops with "Cannot find directory". The version of each call that would have worked, `ggsave("figures/thresholds_indep.pdf", ...)`, sits commented out one line above. The rewrite writes to `maintained/output/` instead.

What the archive never writes. The three `write_rds()` calls at the end of `prep_data.R` are commented out, as are `write(out, "figures/tab_overview.txt")` in `table_2.R`, the two `write()` calls in `table_A2.R` and `write(out, "figures/tab_overview_party.txt")` in `table_A3.R`. A clean exit is easy to mistake for success here: four of six scripts finish without error, and none of them leaves anything behind. `ground_truth/run_deposited_scripts.R` recovers the numbers by sourcing each script unmodified and then writing out the objects it built, which is how `value_script` in the ground truth is filled.

The three deposited `.rds` files are exactly rebuildable. `prep_data.R` derives `dat_clean.rds`, `dat_long_clean.rds` and `long_topic_clean.rds` from `data.csv`, and the deposit ships both the inputs and the outputs. The rewrite rebuilds them rather than loading them, and checks the result with `identical()`, which is what closes the gap the previous paragraph opens: the shipped copies are what the deposited code would have written. The serialized bytes differ, since an `.rds` records the R version that wrote it; the objects do not.

Table 3: Rebuilt from data.csv against the deposit's own copies

Object	Rows rebuilt	Rows deposited	Identical
dat_clean.rds	4,838	4,838	TRUE
dat_long_clean.rds	72,570	72,570	TRUE
long_topic_clean.rds	14,514	14,514	TRUE

Deprecations, none of them fatal. `rm(list = ls())` opens every script. `ifelse()` is used throughout where `if_else()` is now preferred. `table_A2.R` mixes two `%>%` into otherwise native-pipe code, the only two in the deposit, and uses `summarize_at(vars(...))`, superseded by `across()`. `table_A3.R` wraps a tidy data frame in `cbind()` inside `summarize()`, which works but says nothing. `map_df()` is superseded by `map()` plus `list_rbind()`. `summarize()` emits grouping messages throughout. None of these changes a number.

4 Ground Truth Verification

Every published float is covered. `value_paper` was typed from the article, read off rendered pages rather than a text extraction, because a six-column regression table read from a text dump is one column-shift away from a phantom erratum. `value_script` is read out of what the deposited scripts themselves compute, and `value_rewrite` out of `maintained/output/`, so neither can drift from the code behind it.

Table 4: Ground truth coverage by published float

Table or figure	Rows	Match	Miss	Not verifiable
Table 2	61	54	0	7
Table A.2	84	84	0	0
Table A.3	91	84	0	7
Table A.1	38	3	35	0
Figure 1	3	0	2	1
Figure A.1	1	0	0	1
Table 1	1	0	0	1
Text, p. 1219	5	0	2	3
Figure 1 caption	3	0	3	0
Text, p. 1217	2	2	0	0
Text, p. 1222	12	7	5	0
Text, p. 1231	2	2	0	0

The rows that cannot be verified are of three kinds and each is a statement about the article rather than a gap in the check. Fourteen cells across Tables 2 and A.3 carry a p-value the article prints as `p < 0.001` rather than as a number; a single row per table records that all seven such cells in each fall below 0.001 in the rewrite, which they do. Three in-text quantities cannot be rebuilt from a respondent file: two of them, the 6,877 who began the survey and the 73 percent who finished it, count respondents who by construction are not in it, and the third is FiveThirtyEight's Biden approval average, which is not a quantity the survey produces at all. Table 1 prints the two question wordings and no numbers, and the two figures print no numbers either, so what is checked about them is every proportion they plot. Table A.2, which the article calls a numerical version of Figure 1, is not one: the table reports weighted percentages and the figure plots unweighted shares of respondents, and the two differ by up to 3.6 points.

Table 5: Every claim that does not reproduce (47 of 283)

Table or figure	Claim	Paper	Rewrite	Locus

Table A.1	Gender, Male, unweighted (%)	48.10	48.700	archive
Table A.1	Gender, Male, weighted (%)	47.50	47.900	archive
Table A.1	Gender, Female, unweighted (%)	49.90	49.400	archive
Table A.1	Gender, Female, weighted (%)	50.20	49.900	archive
Table A.1	Age, 18-24, weighted (%)	12.10	12.300	archive
Table A.1	Age, 25-34, unweighted (%)	10.30	10.100	archive
Table A.1	Age, 25-34, weighted (%)	17.50	17.200	archive
Table A.1	Age, 35-44, unweighted (%)	13.10	12.900	archive
Table A.1	Age, 35-44, weighted (%)	16.70	16.400	archive
Table A.1	Age, 45-54, unweighted (%)	17.00	16.800	archive
Table A.1	Age, 45-54, weighted (%)	16.00	15.800	archive
Table A.1	Age, 55-64, unweighted (%)	21.00	21.200	archive
Table A.1	Age, 55-64, weighted (%)	16.60	16.800	archive
Table A.1	Age, 65+, unweighted (%)	33.00	33.500	archive
Table A.1	Age, 65+, weighted (%)	21.00	21.500	archive
Table A.1	Race and ethnicity, White, non-Hispanic, unweighted (%)	68.20	69.000	archive
Table A.1	Race and ethnicity, White, non-Hispanic, weighted (%)	63.90	64.800	archive
Table A.1	Race and ethnicity, Black, non-Hispanic, unweighted (%)	11.50	11.600	archive
Table A.1	Race and ethnicity, Black, non-Hispanic, weighted (%)	12.70	12.900	archive
Table A.1	Race and ethnicity, Hispanic, unweighted (%)	12.50	12.000	archive
Table A.1	Race and ethnicity, Hispanic, weighted (%)	16.40	15.800	archive
Table A.1	Race and ethnicity, Asian, non-Hispanic, unweighted (%)	2.70	2.500	archive
Table A.1	Race and ethnicity, Asian, non-Hispanic, weighted (%)	5.40	4.900	archive
Table A.1	Race and ethnicity, Other, non-Hispanic, unweighted (%)	5.00	4.900	archive
Table A.1	Educational attainment, High school or less, unweighted (%)	17.90	17.800	archive
Table A.1	Educational attainment, High school or less, weighted (%)	38.50	38.400	archive
Table A.1	Educational attainment, Some college/associate's, unweighted (%)	31.10	31.000	archive
Table A.1	Educational attainment, Bachelor's, unweighted (%)	28.50	28.400	archive

Table A.1	Educational attainment, Bachelor's, weighted (%)	19.50	19.400	archive
Table A.1	Educational attainment, Graduate degree, unweighted (%)	22.50	22.800	archive
Table A.1	Educational attainment, Graduate degree, weighted (%)	11.50	11.700	archive
Table A.1	Biden approval, Approve, unweighted (%)	43.30	44.200	archive
Table A.1	Biden approval, Approve, weighted (%)	40.40	41.400	archive
Table A.1	Biden approval, Disapprove, unweighted (%)	54.00	55.100	archive
Table A.1	Biden approval, Disapprove, weighted (%)	56.30	57.700	archive
Figure 1	Panel (b) shows Republican primary voters (1 = yes)	1.00	0.000	paper_internal
Figure 1	Panel (c) shows Democratic primary voters (1 = yes)	1.00	0.000	paper_internal
Text, p. 1219	Respondents who completed the survey	5011.00	4838.000	archive
Text, p. 1219	Weighted Biden approval (%)	40.40	41.369	archive
Figure 1 caption	N, respondents intending neither primary	754.00	889.000	paper_internal
Figure 1 caption	N, Republican primary voters	2026.00	2028.000	paper_internal
Figure 1 caption	N, Democratic primary voters	1953.00	1921.000	paper_internal
Text, p. 1222	Republican primary voters, effect on primary vote, p-value	0.02	0.007	paper_internal
Text, p. 1222	Neither primary, effect on belief (pp)	1.80	1.454	paper_internal
Text, p. 1222	Neither primary, effect on belief, p-value	0.05	0.124	paper_internal
Text, p. 1222	Neither primary, effect on general election vote (pp)	0.10	0.279	paper_internal
Text, p. 1222	Neither primary, effect on general election vote, p-value	0.89	0.752	paper_internal

`defect_locus` records where each failure lives. `archive` means the deposit cannot support the claim: here, the 173 respondents it does not contain. `paper_internal` means the deposit and the rewrite agree with each other and with the printed tables, and the article's prose does not.

4.1 In-text claims

The ground truth is organised by published float, which leaves the article's prose thinly covered, and the prose is where three of the four quantitative errors below live. `ground_truth/published_claims.csv` closes that gap. It lists every numeric token in the article and its appendix, 414 of them, each with its location and a hand-assigned type: 348 are quantities this pipeline computes or claims about their shape, 30 are copied from other people's polls and from the American Community Survey, and the rest are question wordings, scale endpoints and dates.

`maintained/in_text_claims.R` carries an entry for each claim the pipeline can reach, with the article's sentence quoted verbatim above the code that computes the number in the article's own units and rounding. It reads only `maintained/output/` and `maintained/clean_data/` and never the ground truth, so the two instruments arrive at each number by separate paths. `build_ground_truth.R` runs it, counts the claim lines it prints, and stops the build unless they account for every claim the extraction says needs one. Counting rather than matching comment markers is the point: an entry that errors part way through prints a prefix of its lines and satisfies a textual check completely.

4.2 Errata

`errata.qmd` builds `barari_etal_2024_errata.pdf`, which records seven errors. Four are places where the article states something its own tables and data do not support: Figure 1's panels are printed in an order the caption and the body text do not describe, page 1222 gives two estimates and two p-values that its own Table 2 contradicts, page 1222 gives a third p-value as 0.02 where Table 2 prints 0.007, and Figure 1's caption reports the counts of respondents by partisan identity rather than by intended primary. Three are typographic, and all three were found by reading the remastered edition against the published pages: page 1217 runs a colon into the word after it, and two reference URLs print a space where a hyphen belongs, so neither is an address a reader can follow. None of the seven changes a conclusion. The corrected values are computed when the document is rendered, so the note cannot go stale the way the article did.

The 173 respondents the deposit does not carry are deliberately absent from that note. A published sentence the archive cannot check is not an erratum: nothing here establishes that 5,011 is the wrong number, only that the deposited file holds 4,838.

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5 Maintained Rewrite

Table 6: Maintained rewrite scripts

Script	Output
helpers.R	Shared packages and the path anchor
clean_data.R	clean_data/*.rds, output/clean_data_vs_deposit.csv
table_a1_demographics.R	output/table_a1_demographics.csv
table_2_avg_effect_by_primary.R	output/table_2_avg_effect_by_primary.csv
table_a2_magnitude_by_primary.R	output/table_a2_magnitude_by_primary.csv
table_a3_avg_effect_by_party.R	output/table_a3_avg_effect_by_party.csv
figure_1_thresholds_by_primary.R	output/figure_1{a,b,c}_*.{pdf,png} and the plotted values as csv
figure_a1_thresholds_by_party.R	output/figure_a1{a,b,c}_*.{pdf,png} and the plotted values as csv
text_sample_and_weighting.R	output/text_sample_and_weighting.csv

`table_a1_demographics.R` has no counterpart in the deposit. It collapses the deposited response options into the categories Table A.1 prints (five race categories, four education bands, approve against disapprove) and computes weighted and unweighted shares on the full-sample denominator. That denominator is why the published Gender and Biden approval columns fall short of 100: a third gender option and a "no answer" on approval sit in it without appearing on the page.

Both figure scripts write a CSV of every proportion they plot. A PDF differs from its predecessor on the timestamp alone, so without that CSV the reproduction check on a figure-only output is vacuous. Those CSVs are also what make the comparison against the deposited scripts possible: all 1,206 plotted proportions in Figure 1 and 1,717 in Figure A.1 agree with the values `figure_1.R` and `figure_A1.R` compute before they fail, to within 5.6e-16, which is floating-point noise rather than a difference in the numbers.

Table 7: Deprecated patterns and replacements

Original	Rewrite
rm(list = ls())	removed
library() in each script	source(helpers.R); all packages in helpers.R
setwd() and bare filenames	here::here()
ggsave() to a hardcoded Windows path	ggsave() to maintained/output/
write() and print.xtable to the console	write_csv() to maintained/output/
commented-out write_rds() calls	write_rds() to maintained/clean_data/
%>%	|>
ifelse()	if_else()
map_df()	map() + list_rbind()
summarize_at(vars(N, Pct), sum)	summarize(across(...)) written out explicitly

cbind(tidy(lm_robust(...)))	reframe(tidy(lm_robust(...)))
theme_minimal()	theme_bw()

What was left alone. Two decisions in the deposit look odd and are not the rewrite's to change. The counterfactual difference is estimated by `lm_robust(value ~ 1, weights = weight)`, an intercept-only weighted regression, which is a weighted mean with an HC2 standard error; it is unusual notation for a familiar quantity but it is the right quantity, and it is what the article reports. And Table A.2's magnitude bands are cut on the absolute difference with `abs_value <= 0.05` counted as "1-5 percent", so a respondent who moved by exactly zero is separated out first and the bands are closed on the right. Both are analytical choices, so both stand.

The archive's `xtable` calls do not survive, but not because `xtable` is unwelcome: the deposit uses it to build a body of LaTeX rows that it never writes to a file, and the rewrite writes tidy CSVs that the ground truth can read cell by cell. The deposited LaTeX is still produced, in the scratch copy, by `ground_truth/run_deposited_scripts.R`, which is what fills `value_script`.

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6 Figure Verification

Figure 1's three panels, in the order the article prints them.

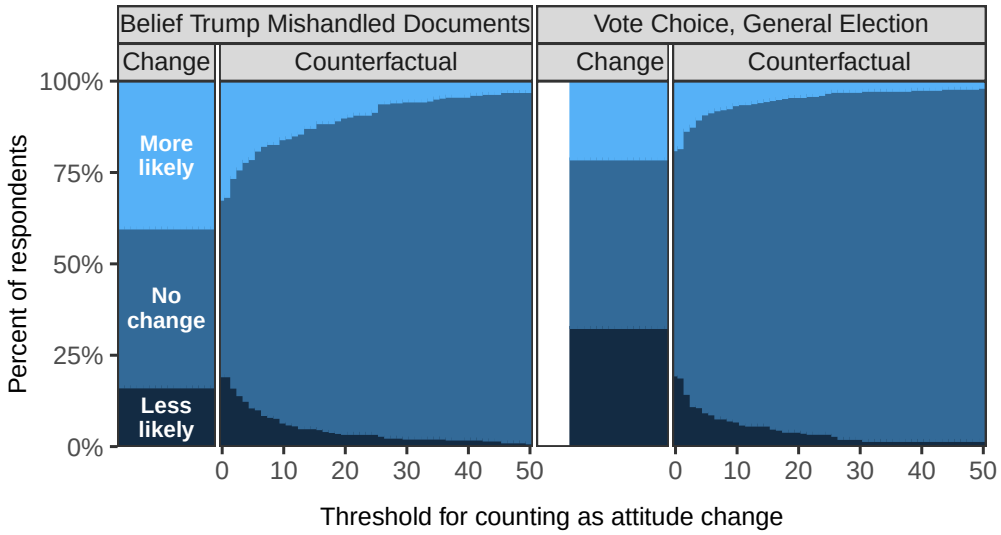


Figure 1: Figure 1a: respondents who intend to vote in neither primary.

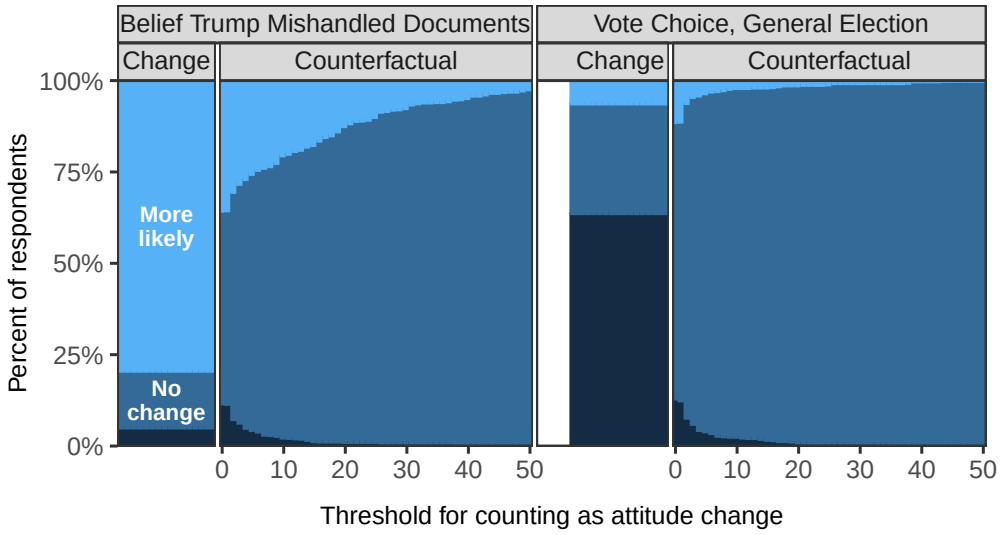


Figure 2: Figure 1b: Democratic primary voters. The caption and body text of the article assign this panel to Republican primary voters.

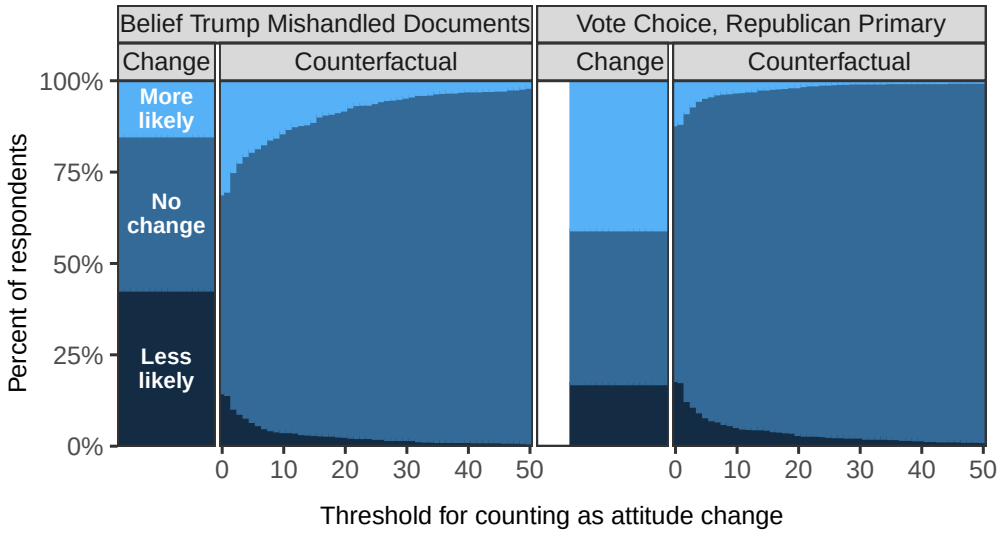


Figure 3: Figure 1c: Republican primary voters. The article's text assigns this panel to Democratic primary voters.

Each panel puts the change format on the left of its own vertical rule and the counterfactual format to the right, swept across thresholds from 0 to 50 points. The comparison the article draws is between the change-format bar and the counterfactual stack at threshold 0, where any

difference at all counts as change. In panel (a) the largest gap between the two formats is 16 points. In panels (b) and (c) it reaches 51 and 29, and the divergence runs in the direction each group's partisanship would predict, which is the evidence for response substitution.

The panel lettering in the article is transposed, as described in the Summary. The maintained files are named for the group each panel actually shows, so `figure_1b_thresholds_democratic` is the file corresponding to the article's printed panel (b).

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7 R Environment

Table 8: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	2.0.1
ggh4x	0.3.1
xtable	1.8.8
knitr	1.51
kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)

The deposit's README records the environment it was built in: R 4.3.1 on Windows, tidyverse 2.0.0, estimatr 1.0.0, ggh4x 0.2.6, xtable 1.8-4, kableExtra 1.3.4, on 7 January 2024.